1. is an Azure Virtual Machine, and how does it work?

- An **Azure Virtual Machine (VM)** is an on-demand, scalable computing resource provided by Microsoft Azure.
- It works by emulating a physical computer, running an operating system (Windows/Linux), and allowing users to run applications, host websites, or process data without managing the underlying hardware.

2. Compare Azure Virtual Machines with on-premises servers:

- **Provisioning:**
  - Azure VMs: Quick setup — can create a VM in minutes.
  - On-Premises Servers: Requires hardware purchase, setup, and installation.
- **Scalability:**
  - Azure VMs: Easily scales up/down based on demand.
  - On-Premises Servers: Scaling involves adding physical servers — slow and costly.
- **Cost:**
  - Azure VMs: Pay-as-you-go or reserved pricing models.
  - On-Premises Servers: High upfront cost for hardware, power, and maintenance.
- **Maintenance:**
  - Azure VMs: Managed by Microsoft — handles hardware failures and updates.
  - On-Premises Servers: Requires manual maintenance and monitoring by IT staff.
- **Accessibility:**
  - Azure VMs: Accessible from anywhere via the internet.
  - On-Premises Servers: Limited to internal networks unless configured for remote access.
- **Disaster Recovery:**
  - Azure VMs: Built-in options like geo-redundancy and backups.
  - On-Premises Servers: Needs custom disaster recovery planning and backup solutions.
- **Security:**
  - Azure VMs: Integrated with Azure Security Center, firewalls, and identity management.
  - On-Premises Servers: Security depends entirely on internal measures.

3. What are the different VM series and their use cases?

- **B-series (Burstable)** — Cost-effective for workloads with unpredictable spikes, like small databases or development environments.
- **D-series (General-purpose)** — Balanced CPU and memory, great for web servers, apps, and enterprise workloads.
- **F-series (Compute-optimized)** — High CPU-to-memory ratio, ideal for batch processing and analytics.
- **E-series (Memory-optimized)** — High memory, used for in-memory databases and big data apps.
- **N-series (GPU-optimized)** — Equipped with GPUs, suitable for AI, deep learning, and graphics rendering.
- **H-series (High-performance)** — For scientific simulations and heavy computations.

4. Explain the pricing model for Azure Virtual Machines:

1. **Pay-as-you-go:**
    a. Charged per second for active VMs — best for short-term, unpredictable workloads.
2. **Reserved Instances:**
    a. Commit to 1 or 3 years for up to 72% cost savings — suitable for stable workloads.
3. **Spot Instances:**
    a. Purchase unused capacity at discounted rates — ideal for non-critical, interruptible workloads.

5. What are the benefits of using Availability Sets and Scale Sets?

- **Availability Sets:**
    - Ensure **high availability** by distributing VMs across fault domains (hardware racks) and update domains (maintenance schedules).
    - Protects against hardware failures and planned maintenance downtime.
- **Scale Sets:**
    - Automatically **scale VMs up or down** based on demand — great for apps with variable traffic.
    - Ensures load balancing and helps handle sudden spikes without manual intervention.

What is Azure App Service, and how does it differ from Virtual Machines?

- **Azure App Service** is a **PaaS (Platform as a Service)** that lets you host web apps, RESTful APIs, and mobile backends without managing the underlying infrastructure.
- It automatically handles deployment, scaling, and load balancing.

List the different types of App Services:

1. **Web Apps:** Host websites and web apps using .NET, Java, PHP, Python, etc.
2. **API Apps:** Build and host RESTful APIs for client consumption.
3. **Mobile Apps:** Backend services for mobile applications, including push notifications and offline sync.
4. **Functions (Serverless):** Execute event-driven code without provisioning servers.

What is auto-scaling in Azure App Services, and how does it help optimize performance?

- **Auto-scaling** automatically increases or decreases the number of app instances based on CPU, memory usage, or HTTP queue length.
- **Benefits:**
    - **Optimizes costs:** Avoid over-provisioning by scaling down when traffic is low.
    - **Improves performance:** Handles traffic spikes by adding instances automatically.

How does Azure App Service integrate with DevOps for Continuous Deployment?

- **Integration:**
    - Connects directly with **GitHub, Azure DevOps, Bitbucket**, or **local Git** repositories.
    - Supports CI/CD pipelines — automatically deploy code changes when new commits are pushed.

What security features are available in Azure App Services?

1. **Managed Identity:** Securely access Azure resources without storing credentials.

2. **Private Endpoints:** Restrict app access to your virtual network, ensuring traffic stays private.
3. **Authentication and Authorization:** Built-in support for OAuth, OpenID Connect, and social logins (Google, Facebook).
4. **TLS/SSL Certificates:** Enforce HTTPS to encrypt data in transit.
5. **Firewall and IP Restrictions:** Control who can access the app based on IP addresses or ranges.

What is an Azure Virtual Machine?

- An **Azure Virtual Machine** is an **on-demand, scalable computing resource** provided by Microsoft Azure.
- It allows you to run virtual servers in the cloud, just like a physical server, but without needing to manage hardware.

Name three use cases for Azure Virtual Machines:

1. **Hosting Applications:** Run custom web apps, APIs, or databases.
2. **Disaster Recovery:** Set up backup environments for on-premises systems.
3. **Development and Testing:** Create isolated environments for testing apps.

What are the different VM sizes available in Azure?

Azure VMs come in various series, tailored to different workloads:

- **B-series:** Cost-effective for burstable workloads (testing environments).
- **D-series:** General-purpose VMs for most apps.
- **F-series:** High-performance computing with faster processors.
- **E-series:** Optimized for memory-intensive apps (databases, analytics).
- **N-series:** Designed for GPU-heavy tasks (AI, ML, graphics).

How do you connect to an Azure VM remotely?

Using Native SSH, Download the SSH key. And run the command in the command prompt :
ssh -i /path/to/your/private-key.pem username@publicIP

What is the difference between Standard HDD, Standard SSD, and Premium SSD in Azure VM storage?

Storage Type	Performance	Use Case	Cost
Standard HDD	Basic	Dev/test workloads	Low
Standard SSD	Moderate	Web servers, moderate apps	Medium
Premium SSD	High (low latency)	High-performance apps, databases	High

What is an Availability Set in Azure?

- An **Availability Set** is a **logical group** of VMs placed across **multiple fault and update domains**.
- It ensures VMs are distributed so hardware failures or updates don't take all VMs down at once — improving **high availability**.

What is Azure Virtual Machine Scale Sets (VMSS)?

- **VM Scale Sets** automatically create and manage a **group of identical VMs**.
- They help scale VMs **in or out** based on demand, ensuring apps handle traffic spikes without manual intervention.

How does Azure handle VM backups?

- Azure uses the **Azure Backup Service** to:
  - Perform **incremental backups** (only changes since the last backup).
  - **Restore VMs** or individual files during disasters.
- You can schedule backups and set retention policies.

Difference between Reserved VM Instances and Pay-as-you-go pricing:

- **Reserved VM Instances:** Pre-purchase VMs for 1 or 3 years at a discounted rate — ideal for predictable workloads.
- **Pay-as-you-go:** Pay only for what you use, billed per second — best for dynamic or short-term workloads.